

Business Problem Statement

NovaTech, a tech products company, wants to analyze its sales performance, regional profitability, and discounting strategies to optimize revenue growth and protect profit margins. Management faces challenges with uneven regional sales distribution, unclear profitability drivers across categories, and potential margin erosion from unmonitored discounting strategies.

You are tasked with analyzing NovaTech's transactional dataset to answer the following overarching business question:

“How can NovaTech leverage transaction data to optimize regional sales channels, control discount strategies, and maximize product category profitability?”

Deliverables

1. **Data Preparation & Modeling (Python):** Clean and transform raw transaction logs, handle missing values (State, Customer Segment), and engineer custom analytical features (Profit Margin %, Discount Category, High Value Order).
2. **Data Analysis (SQL):** Load cleaned records into PostgreSQL and run structured queries to analyze key business questions across revenue channels, regional margins, top product lines, and discount sensitivity.
3. **Visualization & Insights (Power BI):** Develop an interactive dashboard featuring dynamic slicers, KPI card visualizers, and regional/category charts for executive decision-making.
4. **Report & Recommendations:** Deliver a complete project report with data-backed business strategies on discounting limits, stock allocation, and regional expansion.
5. **GitHub Repository:** Provide a structured repository containing all Python ETL scripts (.py/.ipynb), SQL query files (.sql), the cleaned dataset (.xlsx), and the Power BI report file (.pbix).